

Now hold Ae fixed and let B and C meet A ; the angle cAg between curve and tangent vanishes, exactly as in Lemma VII, so the curvilinear areas Abd and Ace coincide with the rectilinear triangles Afd and Age . But these two triangles share the same vertex angle at A along the tangent Ag , so by Lemma V (similar rectilinear figures) their areas are as the squares of the homologous sides, that is $Afd : Age = Ad^2 : Ae^2$. Since $Ad : Ae = AD : AE$ and the original triangles $\triangle ABD$, $\triangle ACE$ stay always proportional to these blown-up areas, the same duplicate ratio descends to them. Therefore the areas $\triangle ABD$ and $\triangle ACE$ are ultimately one to the other as $AD^2 : AE^2$. *Q.E.D.*